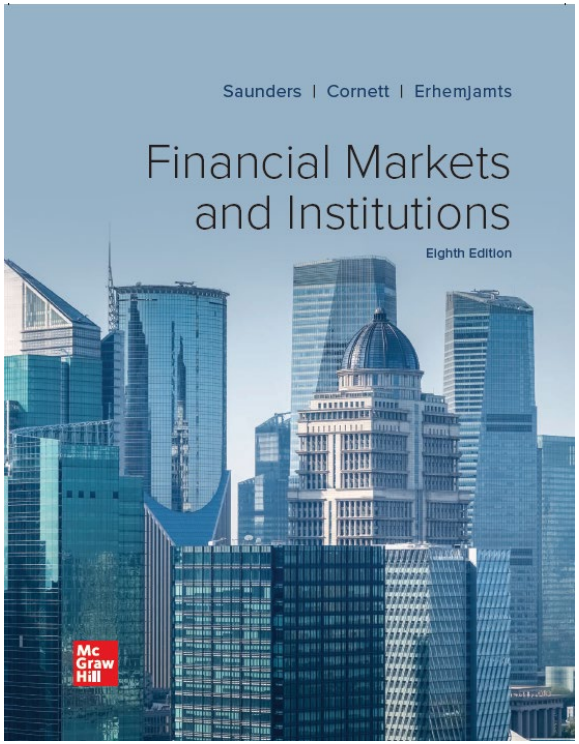
Chapter Five

Money Markets



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Money Markets

- **Money markets** trade debt securities or instruments with maturities of less than one year
  - Once issued, money market instruments trade in active secondary markets
  - Need for money markets arises because the immediate cash needs of individuals, corporations, and governments do not necessarily coincide with their receipts of cash
- Money market instruments share basic characteristics:
  1. Generally sold in large denominations ($1m to $10m units)
  2. Low **default risk**, the risk of late or nonpayment of principal and/or interest
  3. Must have an original maturity of one year or less

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Yields on Money Market Securities

- The *bond equivalent yield* is the quoted nominal, or stated, yield on a security

$$i_{be} = \left[\frac{(P_f - P_0)}{P_0}\right] \times \frac{365}{n}$$

where

P_f = Face value

P_0 = Purchase price of the security

n = Number of days until maturity

$$P_0 = \frac{P_f}{1 + i_{be} \times \frac{n}{365}} \rightarrow \quad i_{be} = \left[\frac{(P_f - P_0)}{P_0}\right] \times \frac{365}{n}$$

- Bond equivalent yield on money market securities with a maturity of less than one year can be converted to an *effective annual interest return* (EAR):

$$EAR = \left(1 + \frac{i_{be}}{365/n}\right)^{365/n} - 1$$

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Yields on Money Market Securities: Discount Yields

- Some money market instruments (e.g., T-bills and commercial paper) are bought and sold on a discount basis
  - Yields on these securities use a 360-day year
  - Return on these securities results from the purchase of the security at a discount from its face value (P_0) and the receipt of face value (P_f) at maturity

- Interest rates on discount securities, or discount yields, are quoted on a discount basis using the following equation:

$$P_0 = \frac{P_f}{1+i_d \times \frac{n}{360}} \rightarrow \quad i_d = \frac{P_f - P_0}{P_f} \times \frac{360}{n}$$

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Money Market Instruments

Treasury bills—short-term obligations issued by the U.S. government.

Federal funds—short-term funds transferred between financial institutions usually for no more than one day.

Repurchase agreements—agreements involving the sale of securities by one party to another with a promise to repurchase the securities at a specified date and price.

Commercial paper—short-term unsecured promissory notes issued by a company to raise short-term cash.

Negotiable certificates of deposit—bank-issued time deposits that specify an interest rate and maturity date and are negotiable (saleable on a secondary market).

Banker's acceptances—time drafts payable to a seller of goods, with payment guaranteed by a bank.

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Money Market Instruments Outstanding

	Amount Outstanding		Yield	
	1990	2019	1990	2019
Treasury bills	$454	$2,515	7.73%	1.57%
Federal funds and repurchase agreements	$351	$4,138	8.29%	1.55%
Commercial paper	$558	$1,131	8.25%	1.77%
Negotiable certificates of deposit	$547	$2,422	8.23%	2.03%
Banker's acceptances	$52	$0	8.00%	—

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Treasury Bills (T-Bills)

- **T-Bills** are short-term debt obligations issued by the U.S. government to cover government budget deficits and to refinance maturing government debt
  - Main tool used by the Federal Reserve to implement monetary policy
  - Original maturities are 4, 13, 26, and 52 weeks
  - Issued in denominations of multiples of $100
  - Existing T-bills can be bought and sold in an active secondary market through government securities dealers who purchase T-bills from the U.S. government and resell them to investors
  - Virtually default risk free, with little interest rate or liquidity risk

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The New Issue and Secondary Market Trading Process for T-Bills

- A **treasury bill auction** is the formal process by which the U.S. Treasury sells news issues of T-bills
  - Bids are submitted by government securities dealers, financial and nonfinancial corporations, and individuals
  - Bids may be *competitive* or *noncompetitive*
  - *Competitive bids* specify the amount of par value of bills desired and the discount yield, rather than the price
    - Generally used by large investors and government securities dealers and make up most of the auction market
  - *Noncompetitive bids* are limited to $5m for a single bidder; they specify only the desired amount of the face value of the bills
    - Usually represent a small portion of total T-bills auctioned, but allow small investors to participate in the T-bill auction market

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Secondary Market for T-Bills

- Secondary market for T-bills is the **largest** of any U.S. money market security
- 24 FIs are designated as ‘primary government securities dealers’ and purchase most of the T-bills sold competitively at auction, creating an active secondary market
  - Around 500 smaller dealers directly trade in this secondary market
  - Primary dealers “make a market” for T-bills by buying and selling securities for their own account and by trading for their customers
  - Secondary market T-bill transactions between primary government securities dealers are conducted over the Federal Reserve’s wire transfer service — Fedwire

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T-Bill Yields

- Discount yield on a T-bill is calculated as follows:
- $P_0 = \frac{P_f}{1+i_d \times \frac{n}{360}} \rightarrow \quad i_{\text{T-bill},\, d} = \frac{P_f - P_0}{P_f} \times \frac{360}{n}$
- The formula for a bond equivalent yield on a T-bill is:
- $P_0 = \frac{P_f}{1+i_{be} \times \frac{n}{360}} \rightarrow \quad i_{\text{T-bill},\, be} = \frac{P_f - P_0}{P_0} \times \frac{365}{n}$

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T-Bill Yield: Example 5-4

EXAMPLE 5–4 **Calculating a Treasury Bill Asked Discount Yield**

Suppose that you purchase the T-bill maturing on March 26, 2020, for $9,961.87. The T-bill matures 90 days after the quote date, December 27, 2019, and has a face value of $10,000. The T-bill's *Asked* discount yield is calculated as:

$$i_{\text{T-bill},\,d} = \frac{\$10{,}000 - \$9{,}961.87}{\$10{,}000} \times \frac{360}{90} = 1.525\%$$

Thus, 1.525 percent is the *Asked* discount yield on this T-bill reported in column 3 of Table 5–5.

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T-Bill Yields (Continued)

- For the asked discount yield, the required market ask price would be:

$$P_0 = P_f - \left(i_{\text{T-bill},\, d} \times \frac{n}{360} \times P_f \right)$$

- For the bond equivalent yield, the required market ask price would be:

$$P_0 = P_f / \left[1 + \left(i_{\text{T-bill},\, be} \times \frac{n}{365} \right) \right]$$

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Federal Funds

- **Federal funds (fed funds)** are short-term funds transferred between financial institutions, usually for a period of one day
  - Commercial banks trade fed funds in the form of excess reserves held at their local Federal Reserve Bank
- **Fed funds rate** is the interest rate for borrowing fed funds
  - Function of the supply and demand for federal funds among financial institutions and the effects of the Federal Reserve's trading through the FOMC
- Primary risk of interbank lending system is that the borrowing bank does not have to pledge collateral for the funds it receives, which are usually in the millions

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Trading in the Federal Funds Market

- Fed funds market is a highly liquid and flexible source of funding for commercial banks and savings banks
  - Commercial banks, especially the largest commercial banks, conduct most transactions in the fed funds market
- **Correspondent banks** are those with reciprocal accounts and agreements
  - Banks with excess reserves call correspondent banks to see if they need overnight reserves, selling the excess reserves to the correspondent banks that offer the highest rates for the fed funds
  - Overnight fed funds loans will likely be based on an oral agreement between the two parties and are generally unsecured loans

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Repurchase Agreements

- A **repurchase agreement (repo or RP)** is an agreement involving the sale of securities by one party to another with a promise to repurchase the securities at a specified price and on a specified date
  - Most repos are collateralized, with the seller holding securities to back the transaction
  - Collateral has a “haircut” applied, meaning the loan is slightly smaller than the market value of securities pledged
  - One-day maturity repos are called *overnight repos*, while repos with longer maturities are called *term repos*
- A **reverse repurchase agreement (reverse repo)** involves the purchase of securities by one party from another with the promise to sell them back

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Commercial Paper

- **Commercial paper** is an unsecured short-term promissory note issued by a company to raise short-term cash, often to finance working capital requirements
  - One of the largest (in terms of dollar value outstanding) of the money market instruments, with nearly $1.1 trillion outstanding as of December 2019
  - Companies with strong credit ratings can generally borrow money at a lower interest rate by issuing commercial paper than by directly borrowing (via loans) from banks
  - Generally sold in denominations of $100,000, $250,000, $500,000, and $1 million
  - Maturities generally range from 1 to 270 days—the most common maturities are between 20 and 45 days
  - Generally held by investors from the time of issue until maturity

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Asset-Backed Commercial Paper (ABCP)

- Mid-2000s saw a huge rise in the use of ABCP
  - In July 2007, $1.19 trillion of the total $2.16 trillion commercial paper outstanding was ABCP
- ABCP is collateralized by other financial assets of issuer
  - The financial assets are typically a mix of many different assets that are jointly judged to have a low risk of bankruptcy by a ratings agency
  - In the mid-2000s, the collateralized assets were mainly mortgage-backed securities
  - Billions of dollars of asset-backed commercial paper were tainted because some of the proceeds were used to buy investments tied to U.S. subprime mortgages
- Even as markets stabilized after the financial crisis, outstanding values of commercial paper continued to fall

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Commercial Paper (Continued)

- Commercial paper is sold to investors either directly using the issuer's own sales force or indirectly through brokers and dealers
  - Commercial paper underwritten and issued through brokers and dealers is more expensive to the issuer
- Like Treasury bills, yields on commercial paper are quoted on a discount basis
- $P_0 = \frac{P_f}{1+i_{cp,d}\times\frac{n}{360}} \rightarrow \quad i_{cp,d} = \frac{P_f - P_0}{P_f} \times \frac{360}{n}$
- To convert to a bond equivalent yield, the following applies:
- $P_0 = \frac{P_f}{1+i_{cp,be}\times\frac{n}{365}} \rightarrow i_{cp,be} = \frac{P_f - P_0}{P_0} \times \frac{365}{n}$

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Negotiable Certificates of Deposit

- A **negotiable certificate of deposit (CD)** is a bank-issued, fixed maturity, interest-bearing time deposit that specifies an interest rate and maturity date and is negotiable
- **Bearer instruments**, meaning whoever holds the CD when it matures receives the principal and interest
- May be traded any number of times in the secondary market
- Denominations range from $100,000 to $10 million; $1 million being the most common
- Often purchased by money market mutual funds with pools of funds from individual investors

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Negotiable Certificates of Deposit (Continued)

- Trading process
  - Banks issuing negotiable CDs post a daily set of rates for the most popular maturities of their negotiable CDs, normally 1, 2, 3, 6, and 12 months
  - Subject to its funding needs, the bank then tries to sell as many CDs to investors who are likely to hold them as investments rather than sell them to the secondary market
- Negotiable CD rates are negotiated between the bank and the CD buyer
  - Large, well-known banks can offer CDs at slightly lower rates than smaller, less well-known banks
  - Negotiable CDs are single-payment securities
  - Interest rates are generally quoted using a 360-day year

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Banker's Acceptances

- A **banker's acceptance (BA)** is a time draft payable to a seller of goods, with payment guaranteed by a bank
  - Make up an increasingly small part of the money markets
  - Beginning in June 2002, banker's acceptance levels are included in the commercial paper levels and no longer differentiated from commercial paper
- Many BAs arise from international trade transactions
- Payable to the bearer at maturity
- Traded in secondary markets
- Denominations determined by the size of the original transaction, but often trade in secondary markets in round lots of $100,000 and $500,000

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Money Market Participants

Instrument	Principal Issuer	Principal Investor
Treasury bills	U.S. Treasury	Federal Reserve System Commercial banks Mutual funds Brokers and dealers Other financial institutions Corporations Individuals
Federal funds	Commercial banks	Commercial banks
Repurchase agreements	Federal Reserve System Commercial banks Brokers and dealers Other financial institutions	Federal Reserve System Commercial banks Mutual funds Brokers and dealers Other financial institutions Corporations
Commercial paper	Commercial banks Other financial institutions Corporations	Brokers and dealers Mutual funds Corporations Other financial institutions Individuals
Negotiable CDs	Commercial banks	Brokers and dealers Mutual funds Corporations Other financial institutions Individuals
Banker's acceptances	Commercial banks	Commercial banks Brokers and dealers Corporations

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